GNANI SAHAYAK (GNANI - VOXORA)

Multilingual Voice-First AI Shopping Assistant for Local Indian Kirana Stores

Powered by Gnani.ai Prisma v2.5 STT & Timbre v2.5 TTS

OFFICIAL GNANI v2.5

132/132 TESTS PASS

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1. THE PROBLEM: VERNACULAR & OPERATIONAL BARRIERS IN KIRANA RETAIL

Local Kirana stores represent over 90% of India's grocery retail, serving hundreds of millions daily.
However, shopkeepers and rural/semi-urban customers face critical operational bottlenecks:

- **Dialectal Code-Switching: Customers fluidly mix vernacular and English (Telugu, Tenglish, Hindi, Hinglish).**
- **Spoken Units & Colloquial Numerals: Quantities are spoken in vernacular terms (ek, do, rendu, aadha, aradha kilo).**
- **Manual Checkout Bottlenecks: Shopkeepers balance weighing goods while writing paper ledgers or typing on clunky POS.**

Existing generic voice bots fail completely on regional accent recognition, Kirana alias mapping, and noise robustness.

2. THE SOLUTION: END-TO-END VERNACULAR VOICE-TO-CART COMMERCE

* **Official Gnani Prisma v2.5 STT:** High-accuracy ASR via api.vachana.ai with concurrency locks & SHA-256 deduplication.
* **Official Gnani Timbre v2.5 TTS:** High-fidelity speech synthesis speaking verified voice receipts in Nalini (HI/EN) & Chitra (TE).
* **5 Language Modes & Auto-Detection:** Instant detection & bidirectional processing for Telugu, Tenglish, Hindi, Hinglish, and English.
* **Vernacular Kirana Alias Engine:** Deterministic matching of colloquial items (e.g. 'biyyam'/'chawal' -> Basmati Rice, 'panchadara' -> Sugar).
* **Conversational Corrections & Steppers:** Hands-free voice modifications ('Change rice to 3 kg') plus tactile UI +/- quantity steppers.
* **Resilient Gateway Architecture:** Integrated HTTP 504/429 backoff retry, HTML sanitization, and preserved audio state on latency.

3. ARCHITECTURE PIPELINE & THREE-SCREEN EXPERIENCE (GNANI - VOXORA)

Screen A: Welcome Threshold

- Language Mode Selection
  (Telugu, Hindi, English, Tenglish)
- Spoken Sample Audio Prompts
- Audio Input Format Validation
  (16 kHz Mono PCM WAV, < 60s)
- **Zero Setup Kirana Entry Point**

Screen B: The Living Bazaar

- Dual Sound Chamber Input
  (Mic recording + WAV upload)
- Live Prisma STT Transcription
- Interactive Grocery Canvas
- Tactile Steppers (+/-) & Pricing
- **Natural Voice Cart Correction**

Screen C: Confirmation Seal

- Itemized Mercantile Receipt Slate
- Subtotal, Tax & Final Grand Total
- Spoken Timbre v2.5 Confirmation
  (Instant audio playback in UI)
- Atomic Inventory Stock Deduction
- **One-Click WhatsApp/Receipt Export**

4. ENGINEERING RIGOR, SECURITY & PRODUCTION READINESS

* **Concurrency Lock Protection:** Internal mutex lock in PrismaClient and TimbreClient blocks overlapping requests and prevents 429 bursts.
* **SHA-256 Audio Deduplication:** Streamlit stateful re-runs never re-bill or re-submit identical audio payloads to Gnani APIs.
* **Resilient Error Sanitization:** Intercepts HTTP 504/502/503 gateway pages; extracts titles, sanitizes HTML, and auto-retries once with backoff.
* **Safe State & Cart Preservation:** Customer cart, order modifications, and recorded WAV audio are preserved intact even upon API latency.
* **Strict Security & Zero Secrets:** API keys loaded strictly via environment variables; credentials masked in logs/exceptions; .gitignore verified.

5. MEASURED BENCHMARKS, VERIFICATION & KIRANA IMPACT

132 / 132 Tests
100% Pytest Passing
Mock + Unit + E2E Suite

3x Faster Checkout
< 15s Order Completion
Hands-free voice billing

5 Dialect Modes
TE, HI, EN, Tenglish, Hinglish
Colloquial & Vernacular ITN

< 1.2s Latency
Gnani Prisma STT + Timbre
Sub-second speech response

SUBMISSION & REPOSITORY VERIFICATION

GitHub Repository: Ready for Public Evaluation with Complete Source, Tests & Demo

Included Components: Streamlit VOXORA UI (app.py) | Gnani Prisma STT & Timbre TTS Clients | Mock & Live Test Harness

Reproducibility: pip install -r requirements.txt && python -m pytest tests/ (132 passed in 2.5s) | Offline Mock Demo Ready

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